Orlando M. Romeo

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EDUCATION

12/2024 **University of California, Berkeley, CA**
Ph.D. – *Doctor of Philosophy,* Earth and Planetary Science

05/2020 **University of Maryland, College Park, MD**
B.S. – *Bachelor of Science,* Physics (magna cum laude)
B.S. – *Bachelor of Science,* Astronomy (magna cum laude)

RESEARCH INTERESTS

Planetary Science:	Crustal magnetization, Planetary magnetospheres, Planetary surfaces
Heliophysics:	Solar wind dynamics, Coronal mass ejections, Wave-particle interactions
Geospace:	Space physics, Space weather prediction, Radiation belt modeling
Instrumentation:	Electrostatic analyzers, X-ray optics, Magnetometers, Space mission design
Data Methods:	Machine learning, Time series analysis, Scientific/numerical modeling

RESEARCH EXPERIENCE

01/2025 – Present **Postdoctoral Researcher**, University of California Berkeley/SSL
Parker Solar Probe (PSP) SWEAP Team, *Advisor: Dr. Davin Larson*

- Support development of data pipelines for electron and ion ESA instruments on PSP
- Generate calibrated science datasets from PSP particle/magnetic field instruments
- Investigate coronal mass ejection evolution using in situ measurements near the Sun
- Optimize 3D ray-tracing algorithms for plasma instrument simulations

FOXSI-5 Sounding Rocket Mission, *Advisor: Dr. Juan Camilo Buitrago-Casas*

- Provide consulting support for the optical alignment system and procedures of the X-ray instruments and white-light camera for the FOXSI-5 sounding rocket

08/2020 – 12/2024 **Graduate Student Researcher**, University of California Berkeley/SSL
InSight/Zhurong Crustal Field Measurements, *Advisor: Dr. Michael Manga*

- Conducted Monte Carlo simulations of crustal magnetic fields on Mars
- Determined probable magnetic coherence scales and depths below each landing site
- Developed geological magnetization models to study the Martian geodynamo history

Parker Solar Probe (PSP) SWEAP Team, *Advisor: Dr. Davin Larson*

- Calibrated the electron ESA instrument on PSP to produce solar wind data products
- Studied electron strahl scattering under varying solar wind conditions
- Collaborated with other instrument teams to produce additional data products

FOXSI-4 Sounding Rocket Mission, *Advisor: Dr. Juan Camilo Buitrago-Casas*

- Assembled, tested, and integrated the Solar Aspect and Alignment System (SAAS)
- Performed laser and X-ray alignment tests and simulations for the optics modules
- Supported multiple stages of rocket integration at SSL, WSMR, & PFRR
- Monitored solar conditions to capture real-time observations of an M-class solar flare

SWFO-L1 STIS Team, *Advisor: Dr. Davin Larson*

- Characterized magnetic emission from the STIS solid-state telescope for SWFO-L1
- Tested Amptek pre-amps on the electronics board for each detector channel
- Performed detector calibration tests with high-energy ion beams and radioactive sources to validate instrument performance
- Monitored STIS conditions during spacecraft Thermal Vacuum Chamber testing

05/2019 – 09/2020 **Planetary Undergraduate Researcher**, NASA Goddard Space Flight Center
MAVEN Proton Cyclotron Waves (PCWs), *Advisor: Dr. Norberto Romanelli*

- Identified proton cyclotron waves (PCWs) in the Martian exosphere using Fourier and minimum variance analyses on MAVEN magnetometer data
- Characterized PCWs and shock processes upstream of the Martian bow shock based on solar wind and planetary properties derived from the MAG and SWIA instruments

08/2017 – 06/2020 **Plasma Physics Undergraduate Researcher**, University of Maryland
Machine Learning Space Weather Forecasting, *Advisor: Dr. Surjalal Sharma*

- Adapted detrending fluctuation analysis on Advanced Composition Explorer (ACE) space weather data to detect long-range correlations over several days
- Forecasted space weather parameters using nonlinear and data-derived techniques (delay embedding, singular value decomposition, and nearest neighbor tree search)

Magnetic Nanoparticle Antennas, *Advisor: Dr. Dennis Papadopoulos*

- Simulated magnetization dynamics for nanoparticles in ferro-fluid antennas from the Stoner-Wohlfarth Model, including effects of magnetization precession and hysteresis
- Calculated nanoparticle magnetization states to determine energy efficient materials for CubeSat antennas in plasma environments

05/2017 – 08/2018 **Space Physics Undergraduate Researcher**, Johns Hopkins University APL
RBSP Data-Driven Forecasting, *Advisor: Dr. Aleksandr Ukhorskiy*

- Modeled radiation belt electron energy flux for each L-shell from Dst index using particle diffusion theory and neural networks
- Filtered electron measurements based on adiabatic invariants from MagEIS and REPT on the Van Allen Probes (RBSP) to calculate Phase Space Density (PSD)
- Forecasted PSD from geomagnetic parameters and geosynchronous satellite observations, implementing nonlinear and data-derived techniques, such as delay embedding, singular value decomposition, and nearest neighbor tree search

01/2017 – 06/2017 **Seismology Undergraduate Researcher**, University of Maryland
Seismic Wave Travel Times of Varying Frequencies, *Advisor: Dr. Vedan Lekic*

- Cross-correlated observed travel times of seismic waves during the 2013 Sea of Okhotsk Earthquake with seismograms from the Preliminary Reference Earth Model (PREM) and Generalized Seismological Data Functionals (GSDF) method
- Adapted Fourier and Gaussian wavelet analyses to filter correlograms based on varying levels of frequency to calculate earthquake phase arrival times
- Developed new framework for earthquake predictions with error bound estimation using significant deviations from PREM to characterize Earth's interior structure

09/2015 – 12/2016 **Space Physics Intern**, Johns Hopkins University APL
Solar Wind Influence on SAA Intensity, *Advisor: Dr. Robert Schaefer*

- Correlated South Atlantic Anomaly (SAA) intensity with ACE solar wind parameters to study space weather influences on geomagnetic activity
- Filtered UV spectrographic imaging from the Defense Meteorological Satellite Program (DMSP) based on photon counts and geographical, lunar and solar positions

TEACHING EXPERIENCE

Academic Courses

University of California Berkeley, Department of Earth & Planetary Science (EPS)

Fall 2021 — **EPS Graduate Student Grader**, *EPS150 – Case Studies in Earth Systems*
- Reviewed weekly manuscripts related to Earth's carbon cycle and Martian hydrology
- Graded weekly student reports on published research papers

Spring 2021 — **EPS Graduate Student Instructor**, *EPS50 – The Planet Earth*
- Instructed students during weekly three-hour lab sections on the study of minerals, rocks, geologic maps, and geological processes
- Designed weekly lab assignments and created questions for midterm and final exams
- Assisted students during weekly three-hour office hours and exam review sessions

University of Maryland, College Park, Department of Astronomy

Fall 2019 — **Astronomy Teaching Assistant**, *ASTR310 – Observational Astronomy*
- Directed weekly two-hour research labs on image processing and data analysis
- Aided students during two-hour office hours with programming troubleshooting
- Graded weekly quizzes, lab activities, exams, and research projects

University of Maryland, College Park, Department of Physics

Spring 2020 — **Physics Teaching Assistant**, *PHYS121 – Fundamentals of Physics I*
- Graded weekly quizzes/exams for 120 students focused on introductory physics
- Held weekly one-hour discussion classes and two-hour office hours
- Led two-hour lab sections using Microsoft Excel, Logger Pro, and physics demos

Fall 2019 — **Physics Teaching Assistant**, *PHYS161– General Physics*
- Graded quizzes/exams for 120 students focused on mechanics and particle dynamics
- Assisted students during weekly two-hour office hours and exam review sessions

Spring 2019 — **Python Session Instructor**, *PHYS205: Developing Essential Research Skills*
- Instructed Python coding sessions over two weeks to undergraduate freshmen
- Provided an overall introduction to Python with basics of coding syntax and logic
- Presented data science methods to visualize and analyze cosmic ray counts from the Bartol Research Institute Neutron Monitor Program

Summer 2018 — **Arduino Program Instructor**, *UMD Physics Summer Girls Outreach Program*
- Led Arduino workshop for 60 high school students to promote women in STEM
- Introduced Arduino code, sensors, and circuits to monitor plant growth over 1 week
- Presented methods to extract and visualize data for plant sustainability statistics

Extracurricular Instruction

Churchville Martial Arts Academy, Churchville MD

01/2016 – 12/2019 — **Martial Arts Head Sensei/Instructor**, *Beginner & Advanced Class*
- Instructed Okinawan Shotokan Karate as a black belt sensei with over 15 years of training to over 200 students aged between 3 and 50 years old
- Organized beginner and advanced classes, belt ranking exams, outreach events, martial arts handbooks and various supplies

Student Mentoring

University of California Berkeley, Space Sciences Lab (SSL)

Summer 2026 — **Heliophysics Mentor**, *Undergraduate Student: Flora Wang*
- Created an automatic detection pipeline for low-energy particle dispersion events observed by PSP and SolO using machine learning

03/2025 – 03/2026 — **Heliophysics Mentor**, *International Undergraduate Student: Yuqi Zhang*
- Developed *CMEMoss*, a public python tool that performs general searches for spacecraft conjunctions with CME trajectories

Summer 2025 — **ASSURE REU Mentor**, *Undergraduate Student: Joseph Byrnes*
- Simulated spacecraft electrostatic charging in various space plasma environments using *SPIS* to support flight calibration and future mission design

Summer 2024 — **FOXSI-5 SAAS/Optics Mentor**, *Graduate Student: Danny Sun*
- Improved the SAAS instrument from previous sounding rocket flights and performed optical alignment tests for the FOXSI-5 sounding rocket mission

Summer 2023 — **ASSURE REU Mentor**, *Undergraduate Student: Elyas Ahmed*
- Utilized machine learning techniques (PySR) to characterize solar wind electron strahl scattering observed by PSP
  - Ahmed, E. *et al.* (2023). Machine Learning for Electron Distributions Observed by PSP. *AGU Fall Meeting Abstracts*, SH31D-3005.

Summer 2022 — **ASSURE REU Mentor**, *Undergraduate Student: Kyla Giron*
- Investigated a potential Earth to Sun connection by studying possible asymmetries in flare and CME production on the Sun
  - Giron, K. *et al.* (2022). Investigating a Possible Earth to Sun Connection - Can the Earth Affect the Sun. *AGU Fall Meeting Abstracts*, ED35D-0579.

University of California Berkeley, Department of Earth & Planetary Science (EPS)

Fall 2023 – Spring 2024 — **EPS Graduate Student Mentor**, *EPS Graduate Mentoring Program*
- Mentored three first-year graduate students in the EPS department
- Established and tracked short/long term goals for students over academic year
- Advised on navigating graduate school, research, and networking

SELECT AWARDS & CERTIFICATIONS

2024	**Certification of Heliophysics Mission Design School** – *NASA/JPL*
2023	**Group Achievement Award for Parker Solar Probe Team** – *NASA*
2023	**Certification of Heliophysics Summer School** – *NASA/UCAR*
2022	**FINESST Fellowship Award (Heliophysics Division)** – *NASA*
2022	**Certification of Solar Orbiter Summer School** – *ESA/CNES*
2022	**SSL Robert P. Lin Graduate Fellowship** – *University of California, Berkeley*
2022	**Travel Award for SHINE 2022 Workshop** – *SHINE*
2019	**William M. MacDonald Physics Scholarship** – *University of Maryland*
2017 – 2020	**Group Award for SPS Outstanding Chapter (Annual)** – *AIS SPS*
2016 – 2020	**JHU APL Academic Merit Scholarship (Annual)** – *JHU APL*
2016 – 2020	**Angelo Bardasis Physics Scholarship (Annual)** – *University of Maryland*

REFEREED PUBLICATIONS

39. Xu, Z., .., **Romeo, O. M.**, *et al.* (2026). Parker Solar Probe Observations of Solar Energetic Particle Events with Inverse-velocity-arrival Features. *Astronomy & Astrophysics*, *710*, A288.
38. Sun, W., .., **Romeo, O. M.**, *et al.* (2026). Parker Solar Probe Observations of Compound Reconnection Exhaust Boundaries and Mirror-mode Structures in the Near-Sun Heliospheric Current Sheet. *The Astrophysical Journal Letters*, *1003(2)*, L26.

37. Huang, Z., ..., **Romeo, O. M.**, *et al.* (2026). Evidence of 5 Minute Oscillations From Parker Solar Probe. *The Astrophysical Journal Letters*, *999(1)*, L4.
36. Goodwill, J., ..., **Romeo, O. M.**, *et al.* (2026). Parker Solar Probe Analysis Across the Alfvénic Transition: Velocity Shear, Magnetic Deflection and Switchback Formation. *Monthly Notices of the Royal Astronomical Society*, stag242.
35. Adhikari, S., ..., **Romeo, O. M.**, *et al.* (2026). Characterization of the Alfvén Transition in the Young Solar Wind Using Parker Solar Probe Observations Approaching Solar Maximum. *The Astrophysical Journal*, *997(2)*, 259.
34. Mozer, F. S., ..., **Romeo, O. M.**, *et al.* (2026). On the Origin of the Dominant Waves in the Extended Solar Corona. *The Astrophysical Journal*, *997(2)*, 355.
33. Zaslavsky, A., **Romeo, O. M.**, Cherier, E., Larson, D. E. (2025). Radial Evolution of the Strahl Pitch-angle Width. *Astronomy & Astrophysics*, *703*, A283.
32. **Romeo, O. M.**, *et al.* (2025). Scales of Martian crustal magnetization constrained by MAVEN, InSight, and Zhurong. *Journal of Geophysical Research: Planets*, *130*, e2025JE008986.
31. Ervin, T., ..., **Romeo, O. M.**, *et al.* (2025). The Impact of Alfvénic Shear Flow on Magnetic Reconnection and Turbulence. *The Astrophysical Journal Letters*, *992(1)*, L11.
30. Pulupa, M., ..., **Romeo, O. M.**, *et al.* (2025). Highly Polarized Type III Storm Observed with Parker Solar Probe. *The Astrophysical Journal Letters*, *987(2)*, L34.
29. Phan, T., **Romeo, O. M.**, *et al.* (2025). Parker Solar Probe Observations of a Highly Energetic and Asymmetric Reconnecting Heliospheric Current Sheet during Encounter 13. *The Astrophysical Journal*, *986(2)*, 209.
28. Huang, J., ..., **Romeo, O. M.**, *et al.* (2025). The Temperature Anisotropy and Helium Abundance Features of Alfvénic Slow Solar Wind Observed by Parker Solar Probe, Helios, and Wind Missions. *The Astrophysical Journal Letters*, *986(2)*, L28.
27. Alnussirat, S. T., ..., **Romeo, O. M.**, *et al.* (2025). Impulsive Solar Flares in the Parker Solar Probe Era. I. Low-energy Electron, Proton, and Alpha Beams. *The Astrophysical Journal Letters*, *985(1)*, 19.
26. Riley, P., ..., **Romeo, O. M.**, *et al.* (2025). Understanding the global structure of the September 5, 2022, coronal mass ejection using sunRunner3D. *Journal of Space Weather and Space Climate*, *15*, 17.
25. Muro, G. D., ..., **Romeo, O. M.**, *et al.* (2025). Radial Dependence of Ion Fluences in the 2023 July 17 Solar Energetic Particle Event from Parker Solar Probe to STEREO and ACE. *The Astrophysical Journal Letters*, *981(1)*, 8.
24. Ruffolo, D., ..., **Romeo, O. M.**, *et al.* (2024). Observed Fluctuation Enhancement and Departure from WKB Theory in Sub-Alfvénic Solar Wind. *The Astrophysical Journal Letters*, *977(1)*, L19.
23. Shaver, S. R., ..., **Romeo, O. M.**, *et al.* (2024). Exploring Observational Heliophysics Across All Scales: Reflections and Insights From the 2023 NASA Heliophysics Summer School. *Perspectives of Earth and Space Scientists*, *5(1)*, e2023CN000217.
22. Ervin, T., ..., **Romeo, O. M.**, *et al.* (2024). Near Subsonic Solar Wind Outflow from an Active Region. *The Astrophysical Journal*, *972(1)*, 129.
21. Phan, T. D., ..., **Romeo, O. M.**, *et al.* (2024). Multiple Subscale Magnetic Reconnection Embedded inside a Heliospheric Current Sheet Reconnection Exhaust: Evidence for Flux Rope Merging. *The Astrophysical Journal Letters*, *971(2)*, L42.
20. Ervin, T., ..., **Romeo, O. M.**, *et al.* (2024). Compositional Metrics of Fast and Slow Alfvénic Solar Wind Emerging from Coronal Holes and Their Boundaries. *The Astrophysical Journal*, *969(2)*, 83.
19. Cohen, C. M. S., ..., **Romeo, O. M.**, *et al.* (2024). Observations of the 2022 September 5 Solar Energetic Particle Event at 15 Solar Radii. *The Astrophysical Journal*, *966(2)*, 148.
18. Zaslavsky, A., ..., **Romeo, O. M.**, *et al.* (2024). Probing Turbulent Scattering Effects on Suprathermal Electrons in the Solar Wind: Modeling, Observations, and Implications. *The Astrophysical Journal*, *966(1)*, 60.
17. Huang, J., ..., **Romeo, O. M.**, *et al.* (2024). Erratum: "Parker Solar Probe Observations of High Plasma β Solar Wind From the Streamer Belt." *The Astrophysical Journal Supplement Series*, *271(2)*, 66.
16. Eriksson, S., ..., **Romeo, O. M.**, *et al.* (2024). Parker Solar Probe Observations of Magnetic Reconnection Exhausts in Quiescent Plasmas near the Sun. *The Astrophysical Journal*, *965(1)*, 76.
15. Palmerio, E., ..., **Romeo, O. M.**, *et al.* (2024). On the Mesoscale Structure of Coronal Mass Ejections at Mercury's Orbit: BepiColombo and Parker Solar Probe Observations. *The Astrophysical Journal*, *963(2)*, 108.

14. McManus, M. D., ..., **Romeo, O. M.**, *et al.* (2024). Proton- and Alpha-driven Instabilities in an Ion Cyclotron Wave Event. *The Astrophysical Journal*, *961(1)*, 142.
13. Mozer, F. S., ..., **Romeo, O. M.**, *et al.* (2023). Density Enhancement Streams in The Solar Wind. *The Astrophysical Journal Letters*, *957(2)*, L33.
12. Alnussirat, S. T., ..., **Romeo, O. M.**, *et al.* (2023). Dispersive Suprathermal Ion Events Observed by the Parker Solar Probe Mission. *The Astrophysical Journal Letters*, *954(1)*, L32.
11. **Romeo, O. M.**, *et al.* (2023). Near-Sun in situ and remote-sensing observations of a Coronal Mass Ejection and its effect on the Heliospheric Current Sheet. *The Astrophysical Journal*, *954(2)*, 168.
10. Huang, J., ..., **Romeo, O. M.**, *et al.* (2023). The Temperature, Electron, and Pressure Characteristics of Switchbacks: Parker Solar Probe Observations. *The Astrophysical Journal*, *954(2)*, 133.
9. Huang, J., ..., **Romeo, O. M.**, *et al.* (2023). The Structure and Origin of Switchbacks: Parker Solar Probe Observations. *The Astrophysical Journal*, *952(1)*, 33.
8. Mozer, F. S., Bale, S. D., Kellogg, P., **Romeo, O. M.**, *et al.* (2023). Arguments for the physical nature of the triggered ion-acoustic waves observed on the Parker Solar Probe. *Physics of Plasmas*, *30(6)*, 062111.
7. Mozer, F. S., ..., **Romeo, O. M.**, *et al.* (2023). Direct observation of solar wind proton heating from in situ plasma measurements. *Astronomy & Astrophysics* , *673*, L3.
6. Huang, J., ..., **Romeo, O. M.**, *et al.* (2023). Parker Solar Probe Observations of High Plasma β Solar Wind from the Streamer Belt. *The Astrophysical Journal Supplement Series*, *265(2)*, 47.
5. Short, B., Malaspina, D. M., Halekas, J., **Romeo, O. M.**, *et al.* (2022). Observations of Quiescent Solar Wind Regions with Near-f_{ce} Wave Activity. *The Astrophysical Journal*, *940(1)*, 45.
4. McManus, M. D., ..., **Romeo, O. M.**, *et al.* (2022). Density and Velocity Fluctuations of Alpha Particles in Magnetic Switchbacks. *The Astrophysical Journal*, *933(1)*, 43.
3. Mozer, F. S., ..., **Romeo, O. M.** (2022). An Improved Technique for Measuring Plasma Density to High Frequencies on the Parker Solar Probe. *The Astrophysical Journal*, *926(2)*, 220.
2. Bandyopadhyay, R., ..., **Romeo, O. M.**, *et al.* (2022). Sub-Alfvénic Solar Wind Observed by the Parker Solar Probe: Characterization of Turbulence, Anisotropy, Intermittency, and Switchback. *The Astrophysical Journal*, *926(1)*, L1.
1. **Romeo, O. M.**, *et al.* (2021). Variability of Upstream Proton Cyclotron Wave Properties and Occurrence at Mars Observed by MAVEN. *Journal of Geophysical Research: Space Physics*, *126(2)*, e28616.

PRESENTATIONS (First Author)

International – Oral

2025 American Geophysical Union Fall Meeting, *Cold Electron Distributions Observed By PSP During the March 13, 2023 CME*, New Orleans, LA, USA

2025 Parker Heliophysics Scholars 10th Meeting, *Cold Electron Distributions Observed By PSP During the March 13, 2023 CME*, Virtual

2024 American Geophysical Union Fall Meeting, *Overview of PSP Electron Observations from 10 to 100 Solar Radii*, Washington D.C., USA

2024 45th COSPAR Scientific Assembly, *Near-Sun In Situ & Remote Sensing Observations of a CME and its Effect on the HCS*, Busan, South Korea

International – Poster

2023 American Geophysical Union Fall Meeting, *Determination and Calibration of Solar Wind Electron Moments Observed by PSP*, San Francisco CA, USA

2023 16th International Solar Wind Conference, *Determining the Anatomy of an ICME by Relating Remote Sensing and In Situ Observations Within 13 Rs*, Pacific Grove CA, USA

2022 American Geophysical Union Fall Meeting, *Characterization of Solar Wind Strahl Electron Scattering Observed by PSP*, Chicago IL, USA

2022 44th COSPAR Scientific Assembly, *Characterization of Strahl Electron Scattering in the Solar Wind Observed by PSP*, Athens, Greece

2022 Solar Orbiter Summer School, *Solar Wind Electron Distributions as Observed by PSP*, Sète, France

2021 American Geophysical Union Fall Meeting, *Characterization of Strahl Electron Scattering in the Solar Wind Observed by PSP*, New Orleans LA, USA

2019 American Geophysical Union Fall Meeting, *Solar Longitudinal Variability of Waves at the Local Proton Cyclotron Frequency*, San Francisco CA, USA

National – Oral

2026 PUNCH 7 Science Meeting, *Parker Solar Probe In situ Measurements of the August 30th, 2025 CME*, Boulder CO, USA

2026 Parker Solar Probe Spring Science Working Group Meeting, *A Comparison of CME Cold Electron Distributions Observed by PSP*, Cambridge MA, USA

2025 JHU/APL Parker Solar Probe Fourth Annual Conference, *Evolution of Electron Distributions During the March 13, 2023, ICME Observed by PSP*, Laurel MD, USA

2023 JHU/APL Parker Sixteenth Science Working Group Meeting, *CME Anatomy from Remote Sensing Observations and In-situ Measurements near 13 Rs*, Pasadena CA, USA

National – Poster

2022 JHU/APL Parker Solar Probe Second Annual Conference, *Characterization of Strahl Electron Scattering in the Solar Wind Observed by PSP*, Laurel MD, USA

2019 AIP/Sigma Pi Sigma Physics Congress, *Seasonal Variability of Waves near the Proton Cyclotron Frequency Upstream from Mars*, Providence RI, USA

2019 MAVEN Project Science Group Meeting, *Seasonal Variability of Waves near the Proton Cyclotron Frequency Upstream from Mars*, Boulder CO, USA

Institutional – Oral

2026 Parker Solar Probe Electron Working Group Meeting, *A Clustering Comparison of CME Cold Electron Distributions Observed by PSP*, Virtual

2025 University of California, Berkeley Center for Integrative Planetary Science Seminar, *Constraining Crustal Coherence Scales of Martian Magnetism*, Berkeley CA, USA

2025 University of California, Berkeley Space Sciences Seminar, *Exploring the Inner Heliosphere with Novel in situ Measurements*, Berkeley CA, USA

2025 Parker Solar Probe Electron Working Group Meeting, *Calibration of the SPAN-E Instruments on Parker Solar Probe*, Virtual

2022 Space Sciences Lab Robert P. Lin Fellowship Seminar, *Electron Distribution Evolution near the Sun Observed by PSP*, Berkeley CA, USA

2018 University of Maryland Astronomical Observatory Open House, *Changes in Brightness for Cataclysmic Variables in Different Galactic Regions*, College Park MD, USA

2018 Johns Hopkins University APL Student Expo, *Data Driven Forecasting Model of Radiation Belt Intensities*, Laurel MD, USA

Institutional – Poster

2019 NASA Goddard Space Flight Center Student Poster Expo, *Seasonal Variability of Waves near the Proton Cyclotron Frequency Upstream from Mars*, Greenbelt MD, USA

2017 University of Maryland Physics Research Showcase, *Seismic Wave Travel Times at Varying Frequencies*, College Park MD, USA

2016 Science and Math Academy Symposium, *Correlation of Solar Wind and IMF Activity on SAA Intensity*, Aberdeen MD, USA

FUNDED GRANTS & CONTRACTS

06/2025 – 03/2031 **Student Electron Electrostatic analyzer (SEE): The Student Collaboration Instrument for HelioSwarm** (Funded by NASA) – Whittlesey (PI), Romeo (Co-I)

09/2025 – 09/2027 **The Superhalo Enigma: Characterizing Energetic Electrons in the Inner Heliosphere** (Funded by ISSI Teams) – Trotta (PI), Romeo (Collaborator)

08/2022 – 08/2025	**Determination of Processes that Control Electron Distribution Evolution near the Sun as Observed by PSP** (Funded by NASA FINESST) – Romeo (FI)

SERVICE & OUTREACH

Professional

2025	NASA ROSES Solicitation, *External Reviewer*
2024	NASA ROSES Solicitation, *Executive Secretary*
2024	The Astrophysical Journal, *Reviewer of 1 manuscript*
2024	Geophysical Research Letters, *Reviewer of 1 manuscript*
2022	The Astrophysical Journal, *Reviewer of 1 manuscript*
2022	Heliophysics 2050 Workshop: Space Physics, *Executive Secretary*
2021	AGU Fall Meeting Parker Solar Probe Session, *Volunteer Chair*
2020	Intersect: The Stanford Journal of STS, *Reviewer of 1 manuscript*

Community

2025 – Present	California Academy of Sciences, *Data Analysis Volunteer*
2025	SSL ASSURE REU Program, *Poster Judge*
2024	California Academy of Sciences SuperNatural Event, *Science Volunteer*
2024	SSL ASSURE REU Program, *Poster Judge*
2023	SSL ASSURE REU Program, *Guest Speaker on Planetary Environments*
2022	SSL ASSURE REU Program, *Student Application Reviewer*
2022, 2023	California Academy of Sciences, *Science Volunteer*
2020	UMD SPS Group Meeting: Python Basics, *Lead Python Workshop Instructor*
2019	UMD Society of Physics Students, *President*
2019	Sigma Pi Sigma Physics Congress, *UMD Booth Volunteer*
2019	UMBC SPS National Zone 4 Meeting, *Lead Python Workshop Instructor*
2018	UMD SPS National Zone 4 Meeting, *Lead Arduino Workshop Instructor*
2017, 2018	UMD Society of Physics Students, *Vice President*

Media Outreach

2024	Science Article Author, *UCB/SSL article on FOXSI-4 rocket launch*
2023	Press Release on *Romeo et al. 2023*, *JHU/APL article and multiple news outlets*

PROFESSIONAL SKILLS

Programming & Software

- Python
- IDL
- MATLAB
- Julia
- Arduino
- Java
- UNIX
- Git
- HTML/CSS

Data Methods

- Data Pipelines
- Data Visualization
- Time Series Analysis
- Signal Processing
- Machine Learning
- Scientific Modeling
- Numerical Simulations
- Data Compression
- Algorithm Development
- Statistical Methods
- Code Optimization
- Parallel Processing

Instrumentation & Lab Experience

- Electrostatic Analyzers
- Magnetometers
- Solid State Telescopes
- Observatory Telescopes
- Oscilloscopes
- Raspberry Pi
- Vacuum Chambers
- 3D Printing
- Circuit Design & Testing
- Optics Alignment
- Magnetic Characterization
- Radioactive Sources

Soft Skills

- Public Speaking
- Community Outreach
- Team Collaborations
- Team Leadership
- Student Mentorship
- Project Management
- Event Coordination
- Scientific Writing

Languages

- English (Native)
- Italian (Proficient)
- Spanish (Beginner)